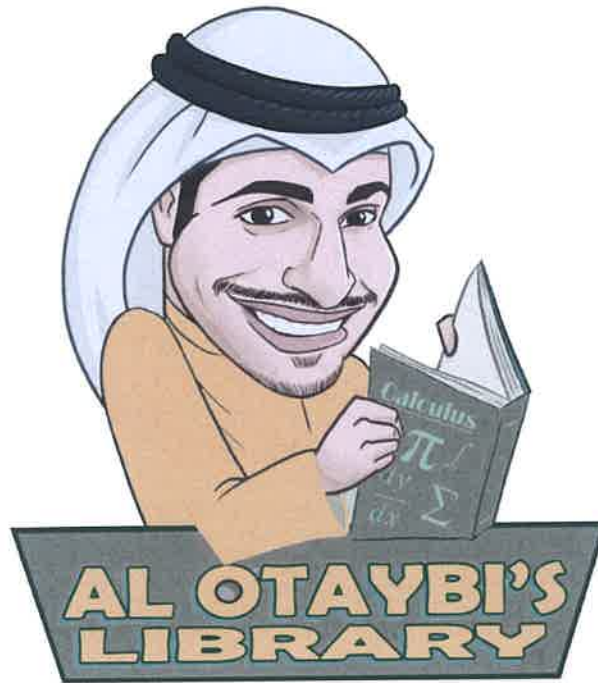


Kuwait University

2021/2022

Spring course



Calculus B

First

Section : (مراجعة)

(1)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi

Review

1) Limit

$$\frac{0}{\text{عدد}} = 0, \quad \frac{\text{عدد}}{\infty} = 0, \quad \frac{\text{عدد}}{0} = \infty$$

الخطوة الأولى من اللمت هي التعويض المباشر

Examples

If $f(x) = \frac{1}{x-5}$

Find (1) $f(5)$ (2) $\lim_{x \rightarrow 5} f(x)$

$$f(5) = \frac{1}{5-5} = \frac{1}{0} = \infty \quad \text{undefind}$$

$$\lim_{x \rightarrow 5} \frac{1}{x-5} = \frac{1}{5-5} = \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow \pm \infty} f(x)$$

Special case when limit approach to infinity

Examples

$$(1) \lim_{x \rightarrow \pm \infty} \frac{3x^2 + 5x}{2x^2 - 9} = \frac{3}{2}$$

$$(2) \lim_{x \rightarrow \pm \infty} \frac{2x}{x^2 + 9} = 0$$

$$(2) \lim_{x \rightarrow \pm \infty} \frac{x^3 + x^2 + 5}{x^2 - 7} = \pm \infty$$



2) Derivative



$$(x^4)' = 4x^3$$

$$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$$

$$\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$$

مشتقة الرقم

بجانبه x



ينزل ثم أشتق x

بروحه



صفر

Examples

Find $f'(x)$

$$f(x) = x^{-2}$$

$$f'(x) = -2x^{-3}$$

$$f(x) = \sqrt{x^5 + x}$$

$$f'(x) = \frac{5x^4 + 1}{2\sqrt{x^5 + x}}$$

$$f(x) = 3 + 3x^5$$

$$f'(x) = 0 + 3(5x^4) = 15x^4$$

$$f(x) = \frac{5}{x}$$

$$f'(x) = -\frac{5}{x^2}$$



طريقة الاشتقاق



(1) هل الدالة مرفوعة لأس؟؟

(2) هل الدالة بسط ومقام؟؟

(3) هل الدالة حاصل ضرب؟؟

Examples

Find $f'(x)$ if

$$f(x) = (x^2 + 1)^5$$

$$f'(x) = 5(x^2 + 1)^4 \cdot 2x$$

$$f(x) = \frac{x^2 + 1}{x^3 - 1}$$

$f(x) =$

المقام

م. البسط

البسط

م. المقام

$(\text{المقام})^2$

$$f'(x) = \frac{(x^3 - 1) \cdot 2x - (x^2 + 1) \cdot 3x^2}{(x^3 - 1)^2}$$

$$f(x) = x^2 \sin x$$

اشتقا بثا + بثا اشتقا

$$f'(x) = 2x \sin x + x^2 \cos x$$



اكسين مضروبين



قوانين الاشتقاق



الدالة	المشتقة
$\ln(f(x))$	$\frac{f'(x)}{f(x)}$
$e^{f(x)}$	$e^{f(x)} \cdot f'(x)$
$a^{f(x)}$	$a^{f(x)} \cdot f'(x) \cdot \ln a$
$\log_a(f(x))$	يعتمد على الأساس a

Find $f'(x)$ if

$$f(x) = \ln(x^5 + x^2)$$

$$f'(x) = \frac{5x^4 + 2x}{x^5 + x^2}$$

$$f(x) = e^{(x^2+5)}$$

$$f'(x) = e^{x^2+5} \cdot 2x$$

$$f(x) = 5^{x^3}$$

$$f'(x) = 5^{x^3} \cdot 3x^2 \cdot \ln 5$$

$$f(x) = \log_2 x$$

$$f(x) = \frac{1}{\ln 2} \ln x$$

$$f'(x) = \frac{1}{\ln 2} \cdot \frac{1}{x}$$

$$f(x) = \log_x \sin x$$

$$f(x) = \frac{1}{\ln x} \ln \sin x$$

$$f(x) = \frac{\ln \sin x}{\ln x}$$

$$f'(x) = \frac{\ln x \cdot \frac{\cos x}{\sin x} - \ln \sin x \cdot \frac{1}{x}}{(\ln x)^2}$$

الدالة	المشتقة
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\csc x$	$-\csc x \cdot \cot x$
$\sec x$	$\sec x \cdot \tan x$
$\cot x$	$-\csc^2 x$

الدالة	المشتقة
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\operatorname{csch} x$	$-\operatorname{csch} x \cdot \coth x$
$\operatorname{sech} x$	$-\operatorname{sech} x \cdot \tanh x$
$\coth x$	$-\operatorname{csch}^2 x$

Find $f'(x)$ if

$$f(x) = \sin \sqrt{x}$$

$$f'(x) = \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

$$f(x) = \cos(x^4)$$

$$f'(x) = -\sin(x^4) \cdot 4x^3$$

$$f(x) = \sin^4 x$$

$$f'(x) = 4 \sin^3 x \cos x$$

$$f(x) = \operatorname{sech}^7 x$$

$$f'(x) = 7 \operatorname{sech}^6 x [-\operatorname{sech} x \tanh x]$$

الدالة	المشتقة
$\sin^{-1}x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1}x$	$-\uparrow$
$\sec^{-1}x$	$\frac{1}{x\sqrt{x^2-1}}$
$\csc^{-1}x$	$-\uparrow$
$\tan^{-1}x$	$\frac{1}{x^2+1}$
$\cot^{-1}x$	$-\uparrow$

Find $f'(x)$ if

$$f(x) = \sin^{-1}(\ln x)$$

$$f'(x) = \frac{\frac{1}{x}}{\sqrt{1-(\ln x)^2}}$$

$$f(x) = \tan^{-1}(x^2)$$

$$f'(x) = \frac{2x}{(x^2)^2 + 1}$$

Find $f'(x)$ if

$$f(x) = e^{x^2} \sec^{-1} x^3$$

اشتقاق e^{x^2} + مشتق $\sec^{-1} x^3$

$$f'(x) = e^{x^2} 2x \sec^{-1} x^3 + e^{x^2} \frac{3x^2}{x^3 \sqrt{(x^3)^2 - 1}}$$

$$f(x) = \frac{\cosh x^2}{\ln x}$$

$$f'(x) = \frac{\ln x \sinh x^2 2x - \cosh x^2 \frac{1}{x}}{(\ln x)^2}$$

$$f(x) = \sinh(x^2) \tan^{-1} x$$

اشتقاق $\sinh(x^2)$ + مشتق $\tan^{-1} x$

$$f'(x) = \cosh(x^2) 2x \tan^{-1} x + \sinh(x^2) \frac{1}{(x)^2 + 1}$$

$$f(x) = \ln(1 + \ln x)$$

$$f'(x) = \frac{1}{1 + \ln x} \cdot \frac{1}{x}$$



3) Integration



$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\int (x+5)^{90} dx$$

$$= \frac{(x+5)^{91}}{91} + c$$

$$\int (9x+2)^{50} dx$$

$$= \frac{(9x+2)^{51}}{51} \times \text{معامل } x$$

الدالة	المشتقة
$\int \frac{1}{x} dx$	$\ln x + c$
$\int e^x dx$	$e^x + c$
$\int a^x dx$	$\frac{a^x}{\ln a} + c$
$\int \frac{1}{\sqrt{1-x^2}} dx$	$\sin^{-1}x + c$
$\int \frac{1}{x^2+1} dx$	$\tan^{-1}x + c$
$\int \frac{1}{x\sqrt{x^2-1}} dx$	$\sec^{-1}x + c$

الدالة	تكاملها	الدالة	تكاملها
$\int \sin x \, dx$	$-\cos x + c$	$\int \sinh x \, dx$	$\cosh x + c$
$\int \cos x \, dx$	$\sin x + c$	$\int \cosh x \, dx$	$\sinh x + c$
$\int \sec^2 x \, dx$	$\tan x + c$	$\int \operatorname{sech}^2 x \, dx$	$\tanh x + c$
$\int \csc x \cdot \cot x \, dx$	$-\csc x + c$	$\int \operatorname{csch} x \cdot \coth x \, dx$	$-\operatorname{csch} x + c$
$\int \sec x \cdot \tan x \, dx$	$\sec x + c$	$\int \operatorname{sech} x \cdot \tanh x \, dx$	$-\operatorname{sech} x + c$
$\int \csc^2 x \, dx$	$-\cot x + c$	$\int \operatorname{csch}^2 x \, dx$	$-\coth x + c$

Find the following integral

$$\int \sin(5x) \, dx$$

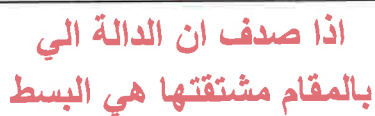
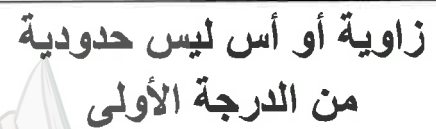
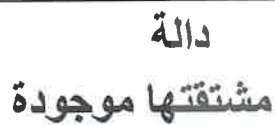
$$= -\frac{\cos(5x)}{5}$$

معامل 5

$$\int \sinh(5x) \, dx$$

$$= \frac{\cosh(5x)}{5} + c$$

معامل 5



عند الفرض

لازم السؤال
يتحول بصيغة
U

لازم المشتقة
تكون موجودة

$$\int \frac{\sin x}{\sqrt{1 - \cos x}} dx$$

$$\int \sin x (1 - \cos x)^{-\frac{1}{2}} dx$$

Let $u = 1 - \cos x$
 $du = \sin x dx$

$$\int u^{-\frac{1}{2}} du$$

$$= \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C = \frac{(1 - \cos x)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$\int \frac{1}{x \sqrt[3]{1 + 2 \ln x}} dx$$

$$\int \frac{1}{x (1 + 2 \ln x)^{\frac{1}{3}}} dx$$

Let $u = 1 + 2 \ln x$
 $du = 2 \frac{1}{x} dx \rightarrow \frac{1}{2} du = \frac{1}{x} dx$

$$\frac{1}{2} \int \frac{1}{u^{\frac{1}{3}}} du$$

$$\begin{aligned} \frac{1}{2} \int u^{-\frac{1}{3}} du &= \frac{1}{2} \frac{u^{\frac{2}{3}}}{\frac{2}{3}} + C \\ &= \frac{1}{2} \cdot \frac{3}{2} (1 + 2 \ln x)^{\frac{2}{3}} + C \end{aligned}$$



زاوية أو أس ليس حدودية
من الدرجة الأولى



الزاوية أو الأس u

Examples

$$\int e^{\sin x} \cos x \, dx$$

Let $u = \sin x$
 $du = \cos x \, dx$

$$\int e^u \, du$$
$$= e^u + C \rightarrow e^{\sin x} + C$$

$$\int e^{4 \tan x} \sec^2 x \, dx$$

Let $u = \tan x$
 $du = \sec^2 x \, dx$

$$\int e^{4u} \, du$$
$$= \frac{e^{4u}}{4} + C = \frac{e^{4 \tan x}}{4} + C$$

$$\int \frac{3^{\sqrt{x}}}{\sqrt{x}} dx$$

Let $u = \sqrt{x}$

$$du = \frac{1}{2\sqrt{x}} dx \longrightarrow 2 du = \frac{1}{\sqrt{x}} dx$$

$$2 \int 3^u du$$

$$= 2 \frac{3^u}{\ln 3} + C = 2 \frac{3^{\sqrt{x}}}{\ln 3} + C$$

$$\int \frac{4^{7 \ln x}}{x} dx$$

Let $u = \ln x$

$$du = \frac{1}{x} dx$$

$$\int 4^{7u} du$$

$$= \frac{4^{7u}}{7 \ln 4} + C = \frac{4^{7 \ln x}}{7 \ln 4} + C$$



دالة
مشتقتها موجودة



الدالة u

Examples

$$\int \frac{\cos x}{1 + \sin^2 x} dx$$

$$\int \frac{\cos x}{1 + (\sin x)^2} dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int \frac{1}{1 + u^2} du = \tan^{-1} u + C$$
$$= \tan^{-1} \sin x + C$$

$$\int \frac{\sec^2 x}{\sqrt{1 - \tan^2 x}} dx$$

$$\int \frac{\sec^2 x}{\sqrt{1 - (\tan x)^2}} dx$$

Let $u = \tan x$
 $du = \sec^2 x dx$

$$\int \frac{1}{\sqrt{1 - u^2}} du = \sin^{-1} u + C = \sin^{-1} \tan x + C$$

$$\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$$

$$\int \frac{e^x}{\sqrt{1-(e^x)^2}} dx$$

Let $u = e^x$

$$du = e^x dx$$

$$\int \frac{1}{\sqrt{1-u^2}} du$$

$$= \sin^{-1} u + C = \sin^{-1} e^x + C$$

$$\int \frac{x}{x^4 + 9} dx$$

$$\int \frac{x}{(x^2)^2 + 9} dx$$

Let $u = x^2$

$$du = 2x dx \longrightarrow \frac{1}{2} du = x dx$$

$$\frac{1}{2} \int \frac{1}{u^2 + 9} du$$

$$\frac{1}{2} \cdot \frac{1}{3} \tan^{-1}\left(\frac{u}{3}\right) + C = \frac{1}{6} \tan^{-1}\left(\frac{x^2}{3}\right) + C$$



إذا صدف ان الدالة الي
بالمقام مشتقتها هي البسط

أسرع سريع

Examples

$$\int \frac{\sec^2 x}{\tan x + 5} dx$$

Let $u = \tan x + 5$
 $du = \sec^2 x dx$

* أسرع سريع *

$$\int \frac{1}{u} du$$
$$= \ln |u| + c$$
$$= \ln |\tan x + 5| + c$$

$$= \ln |\tan x + 5| + c$$

$$\int \frac{\cos x}{\sin x - 2} dx$$

$$= \ln |\sin x - 2| + c$$

$$\int \frac{3\sec^2 x}{\tan x + 5} dx$$

$$= 3 \ln |\tan x + 5| + C$$

$$\int \frac{\sec^2 x}{3\tan x + 5} dx$$

$$\textcircled{\frac{1}{3}} \int \frac{\textcircled{3} \sec^2 x}{3\tan x + 5} dx$$

$$\textcircled{\frac{1}{3}} \ln |3\tan x + 5| + C$$

$$\int \frac{x+5}{x^2+10x} dx$$

$$\textcircled{\frac{1}{2}} \int \frac{\textcircled{2} (x+5)}{x^2+10x} dx$$

$$\textcircled{\frac{1}{2}} \ln |x^2+10x| + C$$



قوانين اضافية للدوال المثلثية

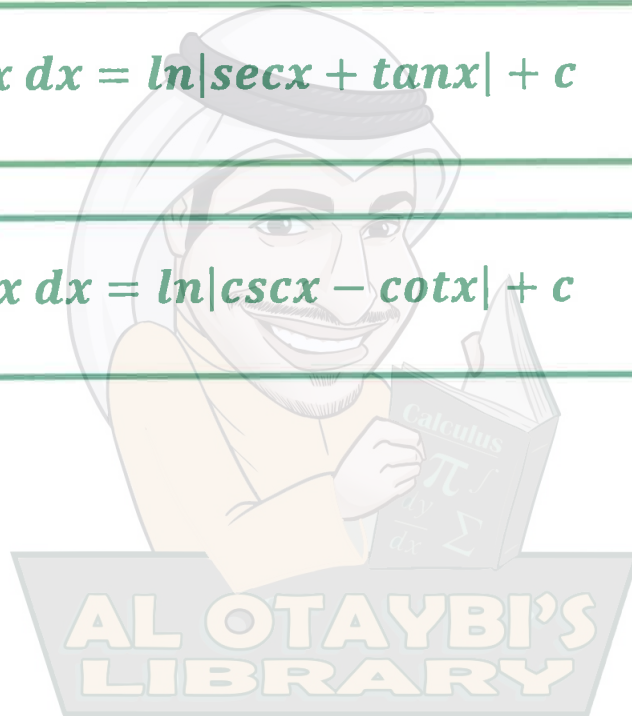


$$\int \tan x \, dx = -\ln|\cos x| + c = \ln|\sec x| + c$$

$$\int \cot x \, dx = \ln|\sin x| + c = -\ln|\csc x| + c$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + c$$

$$\int \csc x \, dx = \ln|\csc x - \cot x| + c$$





Review for angles

مراجعة على الزوايا

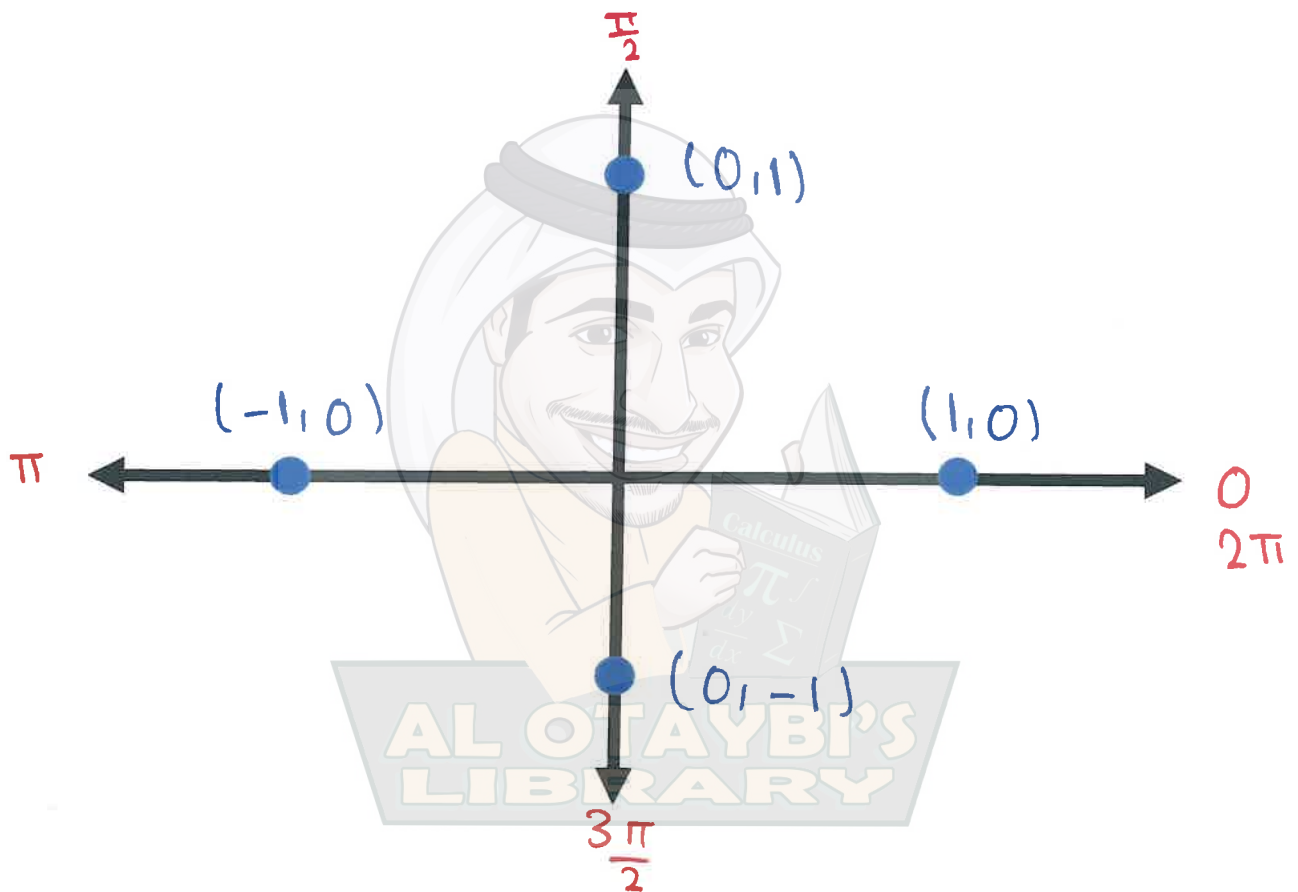
(٤)



الزوايا القائمة



$$P(x, y) = (\cos\theta, \sin\theta)$$



Find the value of

$$\sin(0) = 0$$

$$\tan(0) = \frac{\sin 0}{\cos 0} = \frac{0}{1} = 0$$

$$\cos(0) = 1$$

$$\sec(\pi) = \frac{1}{\cos \pi} = \frac{1}{-1} = -1$$

$$\sin\left(\frac{\pi}{2}\right) = 1$$

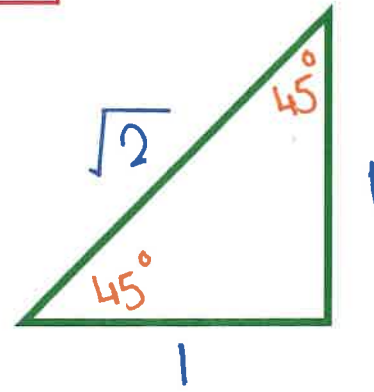
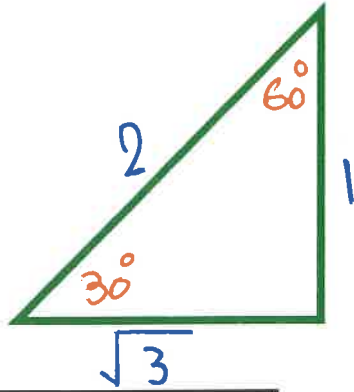
$$\csc\left(\frac{3\pi}{2}\right) = \frac{1}{\sin \frac{3\pi}{2}} = \frac{1}{-1} = -1$$



الزوايا 45° - 60° - 30°



مثلثات



Find the value of

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

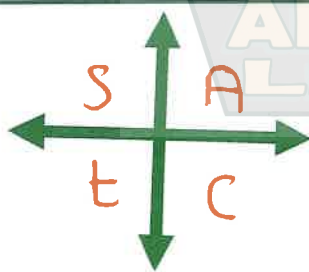
$$\tan\left(\frac{\pi}{4}\right) = \frac{1}{1} = 1$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$\sec\left(\frac{\pi}{4}\right) = \frac{1}{\cos \frac{\pi}{4}} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$



الزوايا الشاذة



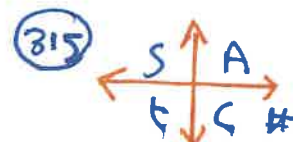
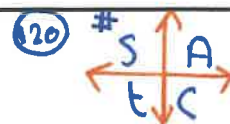
(1) أشوف الربع ← الإشارة

(2) الزاوية = $\frac{\pi}{\text{المقام}}$

Find the value of

$$\sin\left(\frac{2\pi}{3}\right) = + \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\sin\left(\frac{7\pi}{4}\right) = - \sin \frac{\pi}{4} = - \frac{1}{\sqrt{2}}$$





سوالف !!



$\sin^{-1}x$

$$\sin^{-1}(1) = \frac{\pi}{2}$$

$$\sin^{-1}(-1) = -\frac{\pi}{2}$$

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

$$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$$

$$\sin^{-1}(0) = 0$$



$\tan^{-1}x$

$$\tan^{-1}(1) = \frac{\pi}{4}$$

$$\tan^{-1}(-1) = -\frac{\pi}{4}$$

$$\tan^{-1}(0) = 0$$

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$\cos^{-1}x$

$$\cos^{-1}(1) = 0$$

$$\cos^{-1}(-1) = \pi$$

$$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{3\pi}{3} - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\cos^{-1}(0) = \frac{\pi}{2}$$



إذا بطلع الصورة أو اللمت من رسمة

$$\lim_{x \rightarrow \dots} \dots, f(\dots)$$



بمقدار

$x - axis$

(1) أروح حق الـ

(2) (أصعد-أنزل-أثبت) للدالة.

$y - axis$

(3) من نقطة التقاطع , أمد خط أفقي يقطع الـ



الناتج



Zoom

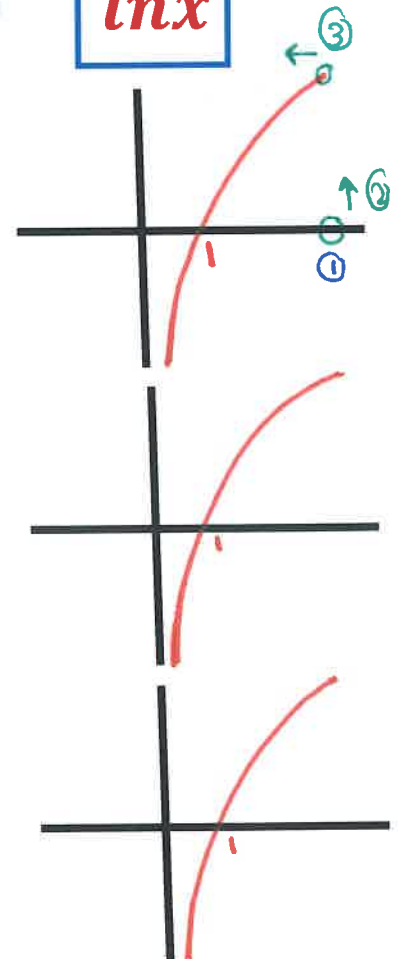
Find the value

$$\lim_{x \rightarrow \infty} \ln x = \ln \infty = \infty$$

$$\lim_{x \rightarrow 0^+} \ln x = \ln 0 = -\infty$$

$$\lim_{x \rightarrow 1} \ln x = \ln 1 = 0$$

$\ln x$





الرسومات , وبعض القيم



$$\ln x$$

$$\log_a x$$

$$\begin{aligned}\ln \infty &= \infty \\ \ln 0^+ &= -\infty \\ \ln 1 &= 0\end{aligned}$$

$$e^x$$

$$a^x$$



$$a > 0$$

$$\begin{aligned}e^\infty &= \infty \\ e^{-\infty} &= 0 \\ e^0 &= 1\end{aligned}$$

$$\tan^{-1} x$$

$$\begin{aligned}\tan^{-1} \infty &= \frac{\pi}{2} \\ \tan^{-1}(-\infty) &= -\frac{\pi}{2} \\ \tan^{-1} 0 &= 0\end{aligned}$$

$$\sinh x$$

$$\cosh x$$

$$\tanh x$$

$$\begin{aligned}\sinh \infty &= \infty \\ \sinh(-\infty) &= -\infty \\ \sinh 0 &= 0\end{aligned}$$

$$\begin{aligned}\cosh \infty &= \infty \\ \cosh(-\infty) &= \infty \\ \cosh 0 &= 1\end{aligned}$$

$$\begin{aligned}\tanh \infty &= 1 \\ \tanh(-\infty) &= -1 \\ \tanh 0 &= 0\end{aligned}$$

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